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BCS-34.

Q#1:-

(a) $\rightarrow$ (i)Soln if $a|b$ and $a|c$, then $a|(b+c)$ Since $a|b$, there is integer $k \Rightarrow b = ka$ $a|c$ there is integer $l \Rightarrow c = la$ So, $b+c = ak+al = a(k+l)$ $\rightarrow$ Since, $(k+l)$ is integer & $(b+c)$ is multiple of a . Hence $a|(b+c)$.

(ii)

Soln if $a|b$ then $a|bc$ $a|b \Rightarrow b = ka$ $a|bc \Rightarrow cb = (ka)c$ $b = a(kc)$ $\therefore kc$ is an integer, because its multiple of a . Pro.

(iii)

if $a|b$ and $b|c$ then $a|c$ $a|b \Rightarrow b = ka \rightarrow (i)$ $b|c \Rightarrow c = bL \rightarrow (ii)$

Substitute (i) in (ii)

 $c = (ka)L$ $c = a(kL)$ hence proved Pro.(b) Soln:- let two integers m & $m+1$. $m(m+1) = m^2 + m \Rightarrow m^2 + m$ $\Rightarrow m = 2n+1$ m is odd. $(2n+1)^2 + (2n+1)$ $= 4n^2 + 4n + 1 + 2n + 1 \Rightarrow 4n^2 + 6n + 2 = 2(2n^2 + 3n + 1)$ $\therefore$ let $m = 2n$ even. $m^2 + m \Rightarrow 4n^2 + 2n \Rightarrow 2n(2n+1)$ $2 \cdot n$ is even.Hence proved $\rightarrow$ The product is always even Pro.

Discrete Assignment 03

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Question #3:

(a) soln:- let's say m & n are ~~even~~ ^{odd} integers then $m+n$ is odd.

$$m = 2k+1$$

$$n = 2l+1$$

$$\begin{aligned} m+n &\Rightarrow 2k+2l \Rightarrow (2k+1) + (2l+1) \Rightarrow 2k+2l+2 \\ &= 2(k+l) \quad \therefore (k+l) \in \mathbb{Z} \\ &= 2\mathbb{Z} \Rightarrow \text{even} \end{aligned}$$

$m+n$ is even & this contradicts with the statement. So, $m+n$ is odd (Original statement is true). $\square$

(b) soln:- for any integer (a) & any prime number (p) if $p|a$ then $p \nmid (a+1)$.

soln:- $\rightarrow$ if $p|a$ then $p|(a+1)$.

$\rightarrow$ if p divides a then it also divides its linear combination.

$$(a+1) - a = 1$$

$$p|1$$

$\rightarrow$ but there exists no prime number which divides 1

So $p|a$ then $p \nmid (a+1)$

$\square$

3(c)

Ans. Set of prime-number is infinite.

R $N = p_1, p_2, p_3, \dots, p_{(n+1)} + 1$

1. $\rightarrow$ None of the listed prime numbers divides N . When dividing N by p_i the remainder should be 1. but there is none which divides N . So, N must be prime itself or shouldn't have prime-factors in the list.

hence there are infinitely many primes.

(ii) Question #4 (a):

$|x| > 1$ then $x > 1$ or $x < -1$ for all $x \in \mathbb{R}$.

Ans.

Contrapositive: If $\neg (x > 1 \text{ or } x < -1)$ then $|x| \leq 1$.

$$\boxed{p \rightarrow q \vee r \\ \neg(q \vee r) \rightarrow \neg p}$$

(b)

$x \leq 1$ and $x \geq -1$

if $-1 \leq x \leq 1$ then $|x| \leq 1$.

$\rightarrow$ its true hence original statement is true. *Ans.*

(ii) (b)

Ans. if $m+n$ is even then m & n are both even or m & n are both odd.

$$m = 2k \quad n = 2l+1$$

$$m+n = 2k + 2l+1$$

$$2(k+l)+1 \therefore k+l = p$$

$$(2p+1) \rightarrow \text{odd}$$

$$\neg p \rightarrow \neg(q \vee r)$$

$$\neg(q \vee r) \rightarrow \neg p$$

$$\neg q \wedge \neg r \rightarrow \neg p$$

(c)

So Contrapositive is true

(c) Ans. Contraposition = ab is not divisible by 4 if a & b are not positive even integers.

$$a = 2k+1$$

$$ab = (2k+1)b = 2kb+b \rightarrow ab \text{ can't contain two factors of 2.}$$

hence statement is true *Ans.*

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Question #5:-

(a) (i) SOL:- for every prime n , $n+1$ is prime.

$$\sum_{n=2}^{\infty} n=23$$

$n=23+1=24 \rightarrow$ not a prime.

(ii) SOL:- Every even divisible by 4.

$$\sum_{n=2}^{\infty} n=10, \frac{10}{4}=2 \rightarrow \text{remainder hence not true.}$$

(b) (i)

SOL:- let $n=10$

counter:- if n is prime then it should only be divisible by itself & 1. so it is false.

(ii) SOL:-

let $n=2$.

$$n^2 > n$$

$$(2)^2 > 2$$

$$4 > 2 \rightarrow \text{True.}$$

(c) let a & b are irrational numbers

so they can't/don't represent in $\frac{p}{q}$ form

$$\text{let } a=\sqrt{2}, b=\sqrt{2}$$

$$\sqrt{2} \times \sqrt{2} = 2 \rightarrow \text{rational.}$$

hence not true.

Question #6 :-

(a)

Soln. $6-7\sqrt{2}$ is irrational
prove by contradiction
let $p = 6-7\sqrt{2}$ a rational number.

$$p-6 = -7\sqrt{2}$$

$$\sqrt{2} = \frac{6-p}{7}$$

→ We get $\sqrt{2}$ which is not a rational. Hence our statement is true. *Pro.*

(c) $4(z^2+z+1)-3z^2$ is a perfect square.

Soln.
let say it's not a perfect square

$$4(z^2+z+1)-3z^2 \Rightarrow z^2+4z+4 = (z+2)^2$$

but $4(z^2+z+1)-3z^2$ can be written as $(z+2)^2$
so our assumption is wrong hence it is a perfect square.

(b) $\sqrt{2} + \sqrt{3}$ is irrational

Soln.

$$\text{let } p = \sqrt{2} + \sqrt{3}$$

$$(p - \sqrt{2}) = (\sqrt{3})^2$$

$$p^2 - 2p\sqrt{2} + 2 = 3$$

$$p^2 - 2p\sqrt{2} = 1$$

$$p^2 - 1 = 2p\sqrt{2}$$

$$\sqrt{2} = \frac{(p^2-1)}{2p}$$

→ $\sqrt{2}$ is irrational hence our statement is true. *Pro.*

Question #7 (a) :-

Soln.

Ben - can $n \leq 1$.

$$1^2 \leq 1 \text{ L.H.S.}$$

$$n = k, 1^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

$$n = k+1$$

$$1^2 + \dots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= (k+1) \left[\frac{k(2k+1) + 6(k+1)}{6} \right]$$

$$= (k+1) \left[\frac{k(2k+1) + 6(k+1)}{6} \right]$$

$$= (k+1) \left[\frac{2k^2 + 7k + 6}{6} \right]$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

Pro.

same as last
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Question #7(b):

Soln.

 $n = k$

$$1^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$$

 $n = k+1$

$$(k+1)^2 \left[\frac{k^2}{4} + (k+1) \right]$$

$$\Rightarrow (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right]$$

$$\Rightarrow (k+1)^2 \cdot \frac{(k+2)^2}{4}$$

$$\Rightarrow \frac{(k+1)^2 (k+2)^2}{4}$$

Ans

Question #7(c):

Soln.

 $n = k$

$$\sum_{i=2}^k \frac{1}{i(i+1)} = \frac{k}{k+1}$$

 $n = k+2$

$$\sum_{i=2}^{k+1} \frac{1}{i(i+1)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

 $\Rightarrow$ Common denominator:-
 $(k+1)(k+2)$

$$= \frac{k(k+2) + 2}{(k+1)(k+2)}$$

$$= \frac{k^2 + 2k + 1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)}$$

$$= \frac{k+1}{k+2} \quad \text{Ans}$$

Question #8(a) :- Soln.

(i) $19/7 = 252, r=5$ Ans

(ii) $-111/10 = 25-10, r=1$

(iii) $789/23 = 2534, r=7$

(iv) $1001/35 = 28, r=0$

(v) $10/19 = 250, r=10$

(vi) $3/5 = 250, r=3$

(vii) $-1/3 = 25-1, r=2$

(viii) $4/5 = 254, r=1$

ie: York
no
is

Question #8 :-

(b)(i) $a = -111$

$m = 99$

Soln.

$a \div m$: quotient

$a \bmod m$: remainder

$$\begin{array}{r} 2 \\ 99 \overline{) -111} \\ \underline{198} \\ 87 \end{array}$$

$q = -2$
 $r = 87$

Ans.

Ans

(ii) $a = -9999$

$m = 101$

Soln.

$$\begin{array}{r} -99 \\ 101 \overline{) -9999} \\ \underline{-1000} \\ 0 \end{array}$$

$q = -99$
 $r = 0$

(iii) $a = 10299$

$m = 999$

Soln.

$q = 10$

$r = 309$

Ans

(iv) $a = 123456, m = 101$

Soln.

$$\begin{array}{r} 123 \\ 101 \overline{) 123456} \\ \underline{123123} \\ 333 \end{array}$$

$q = 123$

$r = 333$ Ans.

Question #8(c) :-

(i) 80

Soln.

$80 \equiv 5 \pmod{17}$

$80 \bmod 17 = 12$

$12 \neq 5$

(Not -Congruent) Ans.

(ii) 103

Soln.

$103 \equiv$

$103 \equiv$

$103 \equiv$

$103 \equiv$

(Not

(iii) -29

Soln.

$-29 \equiv 5 \pmod{17}$

$-29 \bmod 17 = -12$

$-12 + 17 = 5$

(Congruent) Ans

(iv) -122

Soln.

$-122 \equiv 5 \pmod{17}$

$-122 + 119 = -3$

$-3 + 17 = 14$

$-122 \equiv 14 \pmod{17}$

Not congruent to

Ans

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Question #9(a):-

(i) soln.

$$\begin{aligned} \gcd(11, 15) &= 1 \\ \gcd(11, 19) &= 1 \\ \gcd(15, 19) &= 1 \end{aligned}$$

$\Rightarrow$ All are pairwise
have relatively
prime.

(ii) 14, 15, 21

soln.

$$\begin{aligned} \gcd(14, 15) &= 1 \\ \gcd(14, 21) &= 7 \\ \gcd(15, 21) &= 3 \end{aligned}$$

$\Rightarrow$ Not relatively
prime.

(iii) 12, 17, 31, 37

soln.

$$\begin{aligned} \gcd(12, 17) &= 1 \\ \gcd(12, 31) &= 1 \\ \gcd(12, 37) &= 1 \\ \gcd(17, 31) &= 1 \\ \gcd(17, 37) &= 1 \\ \gcd(31, 37) &= 1 \end{aligned}$$

$\Rightarrow$ Relatively-Prime.

(iv) 7, 8, 9, 11

soln.

$$\begin{aligned} \gcd(7, 8) &= 1 \\ \gcd(8, 9) &= 1 \\ \gcd(7, 9) &= 1 \\ \gcd(7, 11) &= 1 \\ \gcd(8, 11) &= 1 \end{aligned}$$

$\Rightarrow$ Relatively Prime.

Question #9(b):-

(i) 88

soln.

$$88 = 2^3 \times 11$$

Ans.

(ii) 126

soln.

$$126 = 2 \times 3^2 \times 7$$

Ans.

(iii) 729

soln.

$$729 = 3^6$$

Ans.

(iv) 1001

soln.

$$1001 = 7 \times 11 \times 13$$

Ans.

(v) 1111

soln.

$$1111 = 11 \times 101$$

Ans.

(vi) 909

soln.

$$909 = 9 \times 101$$

Ans.

Question #9(c) :-

(i) 11, 15, 105

Soln.

11 = 11

15 = 3 × 5

105 = 3 × 5 × 7

GCD = 1

LCM = 11 × 3 × 5 × 7

LCM = 1155 Ans.

(iv) 17, 18, 192, 1021

Soln.

17 = 17

18 = 2 × 3²

192 = 2³ × 3² × 19

1021 = 1021

GCD = 1

LCM = 17 × 3² × 2³ × 19 × 1021

LCM = 9997632 Ans.

(ii) 14, 15, 21

Soln.

14 = 2 × 7

15 = 3 × 5

21 = 3 × 7

GCD = 1

LCM = 2 × 3 × 7 × 5

LCM = 210 Ans.

(iii) 12, 72, 31

Soln.

12 = 2² × 3

72 = 2³ × 3²

31 = 31

32 = 2⁵

GCD = 1

LCM = 2⁵ × 3² × 31

= 8928 Ans.

Question #10(a) :-

(i) GCD(144, 89)

Soln.

as $dx + y$

144 = 89(1) + 55

89 = 55(1) + 34

55 = 34(1) + 21

34 = 21(1) + 13

21 = 13(1) + 8

13 = 8(1) + 5

8 = 5(1) + 3

5 = 3(1) + 2

3 = 2(1) + 1

2 = 2(1) + 0

↓
gcd.

Extended :-

1 = 3 - 1 × 2 as 2 = 5 - 1 × 3 So,

1 = 3 - (5 - 3) ⇒ 2 - 3 + 5

1 = 2(8 - 5) - 5 ⇒ 2 × 8 - 3 × 5

1 = 2 × 8 - (3)(3 - 8) ⇒ 5 × 8 - 3 × 13

1 = 5(21 - 13) - 3 × 13 ⇒ 5 × 21 - 8 × 13

1 = 5 × 21 - 8(34 - 21) ⇒ 13 × 21 - 8 × 34

1 = 13(55 - 34) - 8 × 34 ⇒ 13 × 55 - 21 × 34

1 = 13 × 55 - 21(89 - 55) ⇒ 34 × 55 - 21 × 89

1 = 34(144 - 89) - 21 × 89 = 34 × 144 - 55 × 89

1 = 34 × 144 - 55 × 89

So, 34 × 144 - 55 × 89

Ans

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Question #10(a) :-

(ii) GCD (1001, 10001)

Sol.

$$100001 = 1001 \cdot 99 + 902$$

$$1001 = 902(1) + 99$$

$$902 = 99 \times 9 + 11$$

$$99 = 11 \times 9 + 0$$

↓
gcd

Extended :-

$$11 = 902 - 9 \cdot 99$$

$$11 = 902 - 9(1001 - 902)$$

$$11 = 902 - 9 \cdot 1001 + 9(902)$$

$$11 = 10 \cdot 902 - 9 \cdot 1001$$

$$11 = -9 \cdot 1001 + 10(100001 - 990 \cdot 1001)$$

$$11 = -9 \cdot 1001 + 10 \cdot 100001 - 990 \cdot 1001$$

$$11 = 10 \cdot 100001 - 999 \cdot 1001$$

$$11 = (-999) \cdot 1001 + 10 \cdot 100001$$

Ans

Question #10(b) :-

(i) $55x \equiv 34 \pmod{89}$

Sol.

gcd(55, 89) = 1

$$89 = 55(1) + 34$$

$$55 = 34(1) + 21$$

$$34 = 21(1) + 13$$

$$21 = 13(1) + 8$$

$$13 = 8(1) + 5$$

$$8 = 5(1) + 3$$

$$5 = 3(1) + 2$$

$$3 = 2(1) + 1$$

$$2 = 1(2) + 0$$

↓
gcd

Extended :-

$$1 = 3 - 1 \cdot 2, \quad 2 = 5 - 1 \cdot 3$$

$$1 = 3 - (5 - 3) = 2 \cdot 3 - 5 = 8 - 1 \cdot 5$$

$$1 = 2(8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5 = 13 - 1 \cdot 8$$

$$1 = 2 \cdot 8 - 3(13 - 8) = 5 \cdot 8 - 3 \cdot 13 = 12 - 1 \cdot 13$$

$$1 = 5(21 - 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13 = 34 - 1 \cdot 21$$

$$1 = 13(55 - 34) - 8 \cdot 34 = 13 \cdot 55 - 21 \cdot 34 = 89 - 1 \cdot 89$$

$$1 = 13 \cdot 55 - 21(89 - 55) = 34 \cdot 55 - 21 \cdot 89$$

$$1 = 34 \cdot 55 - 21 \cdot 89$$

So, Inverse of 55 mod 89 is 34

$$34 \cdot 55 \equiv 1 \pmod{89}$$

$$55x \equiv 34 \text{ by } 55^{-1} = 34$$

$$x \equiv 34 \cdot 34 \pmod{89}$$

$$89 \cdot 13 = 1157 \Rightarrow 1156 \equiv -1 \equiv 88 \pmod{89}$$

$$x \equiv 88 \pmod{89}$$

Ans

(ii) $89x \equiv 2 \pmod{232}$

Soln:-
 $232 = (2)89 + 54$
 $89 = 54(1) + 35$
 $54 = 35(1) + 19$
 $35 = 19(1) + 16$
 $19 = 16(1) + 3$
 $16 = 3(5) + 1$
 $3 = 3(1) + 0$
 $\downarrow$
 gcd.

Extended:-

$1 = 16 - 5 \cdot 3$, $3 = 19 - 1 \cdot 16$
 $1 = 16 - 5(19 - 16) = 6 \cdot 16 - 5 \cdot 19 = 35 - 1 \cdot 19$
 $1 = 6(35 - 19) - 5 \cdot 19 = 6 \cdot 35 - 11 \cdot 19 = 54 - 1 \cdot 35$
 $1 = 6 \cdot 35 - 11(54 - 35) = 17 \cdot 35 - 11 \cdot 54 = 89 - 1 \cdot 54$
 $1 = 17(89 - 54) - 11 \cdot 54 = 17 \cdot 89 - 28 \cdot 54 = 232 - 28 \cdot 54$
 $1 = 17 \cdot 89 - 28(232 - 2 \cdot 89) = (17 + 56) \cdot 89 - 28 \cdot 232$
 $1 = 73 \cdot 89 - 28 \cdot 232$
 $73 \cdot 89 \equiv 1 \pmod{232}$
 $x \equiv 73 \cdot 2 \pmod{232}$

Question #10(c) :-

(i) $a=2, m=17$
 Soln

$17 = 8 \cdot 2 + 1$, $2 = 2 \cdot 1 + 0$
 So, inverse $2 \pmod{17}$
 $-8 \equiv 9 \pmod{17}$
 $2^{-1} \equiv 9 \pmod{17}$ Ans

(iii) $a=144, m=233$
 Soln

$1 = 89 \cdot 144 - 55 \cdot 233$
 $1 = 89 \cdot 144 + (-55) \cdot 233$

So, inverse of $144 \pmod{233}$ is 89 .

$144^{-1} \equiv 89 \pmod{233}$

Ans.

(ii) $a=34, m=89$

Soln

Directly written

$1 = (-34) \cdot 34 + 13 \cdot 89$
 $x = -34, y = 13$
 $-34 \cdot 34 + 13 \cdot 89 = 1$

$\therefore -34 \cdot 34 \equiv 1 \pmod{89}$ Ans

(iv) $a=200, m=1001$
 Soln

Extended Euclidean.

$1 = (-5) \cdot 200 + 1 \cdot 1001$

Inverse of $200 \pmod{1001}$ is:-

$-5 \equiv 996 \pmod{1001}$

$200^{-1} \equiv 996 \pmod{1001}$

Ans

16
90/63 35-1
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Question #11(a):

(i) Soln. 125973128.

$$S = (1)(1) + 2(2) + 8(3) + 9(4) + 7(5) + 3(6) + (1)(7) + 2(8) + 8(9)$$

$$S = 1 + 4 + 15 + 36 + 35 + 18 + 7 + 16 + 72$$

$$S = 204 \text{ mod } 11$$

check-digit = 6 Ans

(ii) Soln.

$$7 \times 15 = 105$$

$$105 \text{ mod } 26 = 105 - 26 \times 4$$

$$105 - 104 = 1$$

$$7 \times 15 \equiv 1 \pmod{26} \quad \text{Yes it is Ans.}$$

Question #11(b):

(i) $n = 11 \cdot 3 = 33 \quad e = 3$

Soln

Enam (if we start from a at position zero)

$$e = 4, n = 23, a = 0, m = 12 \rightarrow \text{positions.}$$

Encrypt:-

$$e = 4 \Rightarrow 4^3 \cdot 64 \Rightarrow 64 \text{ mod } 33 = 31$$

$$n = 23 \Rightarrow 23^3 \cdot 12167 \Rightarrow 12167 \text{ mod } 33 = 23$$

$$a = 0 \Rightarrow 0^3 = 0$$

$$m = 12 \Rightarrow 12^3 \text{ mod } 33 = 12 \quad \text{Ans.}$$

(ii) soln

Encryption: $c = (p + 5) \bmod 26$, decryption: $p = (c - 5) \bmod 26$

$$x = 23 \Rightarrow 23 - 5 = 18 \rightarrow S$$

$$y = 24 \Rightarrow 24 - 5 = 19 \rightarrow T$$

$$7 = 19 \Rightarrow 19 - 5 = 14 \rightarrow O$$

$$4 = 20 \Rightarrow 20 - 5 = 15 \rightarrow P$$

$$Y = 24 - 5 = 19 \rightarrow T$$

$$F = 5 - 5 = 0 \rightarrow A$$

$$Q = 16 - 5 = 11 \rightarrow L$$

$$P = 15 - 5 = 10 \rightarrow K$$

$$N = 13 - 5 = 8 \rightarrow I$$

$$S = 18 - 5 = 13 \rightarrow N$$

$$L = 11 - 5 = 6 \rightarrow G$$

So, XYTU YFQPNSL

$\Rightarrow$ Stop Talking

STOP TALKING Ans

Question #11(c) ::

(i) $x \equiv -1 \pmod{25} \rightarrow x = 25k - 1$

$$-100 \leq x \leq 100$$

soln

$$k = 0 \rightarrow -1$$

$$k = 1 \rightarrow 24$$

$$k = 2 \rightarrow 49$$

$$k = 3 \rightarrow 74$$

$$k = 4 \rightarrow 99$$

$$k = -1 \rightarrow -26$$

$$k = -2 \rightarrow -51$$

$$k = -3 \rightarrow -76$$

$$\Rightarrow \{-76, -51, -26, -1, 24, 49, 74, 99\}$$

Ans

(ii) soln

By fermat: $5^{18} \equiv 1 \pmod{59}$

factor of 119 $\rightarrow 58 \times 2 + 3$

So,

$$5^{119} \equiv (5^{58})^2 \cdot 5^3 \equiv 1 \cdot 125 \pmod{59}$$

$$125 \bmod 59 = 7$$

Remainder

Ans

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Question #12(a)

(i) Soln:-

$$x \equiv 1 \pmod{5}, x \equiv 2 \pmod{6}, x \equiv 3 \pmod{7}.$$

$$M = 5 \cdot 6 \cdot 7 = 210$$

$$M_1 = \frac{210}{5} = 42 \quad M_2 = \frac{210}{6} = 35 \quad M_3 = \frac{210}{7} = 30$$

$$a_1 = 1, a_2 = 2, a_3 = 3.$$

So, for (y):

$$42y_1 \equiv 1 \pmod{5} \Rightarrow y_1 = 3$$

$$35y_2 \equiv 1 \pmod{6} \Rightarrow y_2 = 5$$

$$30y_3 \equiv 1 \pmod{7} \Rightarrow y_3 = 4$$

Soln:-

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$$

$$x = 1 \cdot 42 \cdot 3 + 2 \cdot 35 \cdot 5 + 3 \cdot 30 \cdot 4$$

$$126 + 350 + 360 = 836$$

$$836 \pmod{210}$$

$$= 836 - 630 = 206.$$

$$x \equiv 206 \pmod{210}$$

Ans

LWB

Question #12 (a)

(ii) Soln

$$x \equiv 1 \pmod{2}, x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 4 \pmod{11}$$

$$M = 2 \cdot 3 \cdot 5 \cdot 11 = 330$$

$$M_1 = \frac{330}{2} = 165, M_2 = \frac{330}{3} = 110, M_3 = \frac{330}{5} = 66, M_4 = \frac{330}{11} = 30$$

$$a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 4$$

for y:

$$165y_1 \equiv 1 \pmod{2} \rightarrow y_1 = 1$$

$$110y_2 \equiv 1 \pmod{3} \rightarrow y_2 = 2$$

$$66y_3 \equiv 1 \pmod{5} \rightarrow y_3 = 1$$

$$30y_4 \equiv 1 \pmod{11} \rightarrow y_4 = 7$$

$$x = 1 \cdot 165 \cdot 1 + 2 \cdot 110 \cdot 2 + 3 \cdot 66 \cdot 1 + 4 \cdot 30 \cdot 7$$

$$x = 1643 \pmod{330}$$

$$= 323 \text{ Ans}$$

Q#12(b):- $N \equiv 3 \pmod{5}, N \equiv 3 \pmod{6}, N \equiv 1 \pmod{7}, N \equiv 0 \pmod{11}$

$$N \equiv 3 \pmod{\text{LCM}(5,6)=30}$$

Soln. $30k + 3 \equiv 1 \pmod{7} \Rightarrow 30k \equiv -2 \equiv 5 \pmod{7}$

$$30 \equiv 2 \pmod{7} \Rightarrow 2k \equiv 5 \pmod{7}$$

$$\text{So, } k = 7k + 6$$

$$N = 30(7k + 6) + 3 = 210k + 183$$

$$N \equiv 0 \pmod{11}$$

$$210k + 183 \equiv 0 \pmod{11}$$

$$210 \equiv 1 \pmod{11}$$

$$183 \equiv 7 \pmod{11}$$

$$\text{So, } k + 7 \equiv 0 \Rightarrow k \equiv 4 \pmod{11}$$

$$k = 11s + 4$$

$$N = 210(11s + 4) + 183 = 2310s + 840 + 183 = 2310s + 1023$$

Sol. 1023

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⑨

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Question #12(c):

Soln:-

$$P \equiv 3 \pmod{4}, P \equiv 5 \pmod{9}, P \equiv 8 \pmod{13}$$

$$P \equiv 3 \pmod{4}, P \equiv 5 \pmod{9} \quad P = 4a + 3$$

$$4a + 3 \equiv 5 \pmod{9} \rightarrow 4a \equiv 2 \pmod{9}$$

$$a = 9t + 5 \quad \text{so, } P = 4(9t + 5) + 3 = 36t + 23$$

$$P \equiv 8 \pmod{13}$$

$$36t + 23 \equiv 8 \pmod{13}$$

$$36t - 15 \equiv 11 \pmod{13}$$

$$36 \equiv 10 \pmod{13}$$

$$\text{So, } 10t \equiv 11 \pmod{13}$$

$$t \equiv 13s + 5$$

$$P = 36(13s + 5) + 23 \Rightarrow 468s + 180 + 23$$

$$= 468s + 203$$

$$\boxed{P = 203} \quad \text{Ans}$$

Question #13(a):

(i) Shift of 3: $f(p) = (p+3) \bmod 26$.

Sol.

$I \rightarrow 8 + 3 \rightarrow L$	$D \rightarrow 3 + 3 \rightarrow G$	$M \rightarrow 12 + 3 \rightarrow P$
$L \rightarrow 11 + 3 \rightarrow O$	$I \rightarrow 8 + 3 \rightarrow L$	$A \rightarrow 0 + 3 \rightarrow D$
$O \rightarrow 14 + 3 \rightarrow R$	$S \rightarrow 18 + 3 \rightarrow V$	$T \rightarrow 19 + 3 \rightarrow W$
$V \rightarrow 21 + 3 \rightarrow Y$	$C \rightarrow 2 + 3 \rightarrow F$	$H \rightarrow 7 + 3 \rightarrow K$
$E \rightarrow 4 + 3 \rightarrow H$	$R \rightarrow 17 + 3 \rightarrow U$	$E \rightarrow 4 + 3 \rightarrow H$
	$E \rightarrow 4 + 3 \rightarrow H$	$M \rightarrow 12 + 3 \rightarrow P$
	$T \rightarrow 19 + 3 \rightarrow W$	$A \rightarrow 0 + 3 \rightarrow D$
	$E \rightarrow 4 + 3 \rightarrow H$	$T \rightarrow 19 + 3 \rightarrow W$
		$I \rightarrow 8 + 3 \rightarrow L$
		$C \rightarrow 2 + 3 \rightarrow F$
		$S \rightarrow 18 + 3 \rightarrow V$

2) LORYH GLVFUHHH

PDWKHPDWLFV Ans

(ii)

Sol.
(i)

$S \rightarrow 18 + 4 \rightarrow W$	$P \rightarrow 15 + 4 \rightarrow T$
$T \rightarrow 19 + 4 \rightarrow X$	$O \rightarrow 14 + 4 \rightarrow S$
$O \rightarrow 14 + 4 \rightarrow S$	$L \rightarrow 11 + 4 \rightarrow P$
$P \rightarrow 15 + 4 \rightarrow T$	$L \rightarrow 11 + 4 \rightarrow P$
	$U \rightarrow 20 + 4 \rightarrow Y$
	$T \rightarrow 19 + 4 \rightarrow X$
	$I \rightarrow 8 + 4 \rightarrow M$
	$O \rightarrow 14 + 4 \rightarrow S$
	$N \rightarrow 13 + 4 \rightarrow R$

=> WXST TSPPYXMSR

Ans.

(ii) $f(p) = (p+21) \bmod 26$

Sol.

$S \rightarrow N$
$T \rightarrow O$
$O \rightarrow J$
$P \rightarrow K$
$P \rightarrow K$
$O \rightarrow J$
$L \rightarrow G$
$L \rightarrow G$
$U \rightarrow P$
$T \rightarrow O$
$I \rightarrow D$
$O \rightarrow J$
$N \rightarrow I$

=> NOJK KJGGRODI

Ans

Name: Yash-Raj

Roll-no: 24K-0737.

BCS-34.

Question #13(b):

(i) PLG WZR DVVLTJQPHQW

Shifts 3 back:-

P $\rightarrow$ ML $\rightarrow$ IG $\rightarrow$ DW $\rightarrow$ TZ $\rightarrow$ WR $\rightarrow$ OD $\rightarrow$ AV $\rightarrow$ SV $\rightarrow$ SL $\rightarrow$ IJ $\rightarrow$ GQ $\rightarrow$ NP $\rightarrow$ MH $\rightarrow$ EQ $\rightarrow$ NW $\rightarrow$ T(ii) IDVW QXFHVXQLYHU
Shifts 3 back:-I $\rightarrow$ FD $\rightarrow$ AV $\rightarrow$ SW $\rightarrow$ TQ $\rightarrow$ NX $\rightarrow$ UF $\rightarrow$ CH $\rightarrow$ EV $\rightarrow$ SX $\rightarrow$ UQ $\rightarrow$ NL $\rightarrow$ IY $\rightarrow$ VH $\rightarrow$ EU $\rightarrow$ RV $\rightarrow$ SL $\rightarrow$ IW $\rightarrow$ TB $\rightarrow$ Y

e) MID TWO ASSIGNMENT

M.

e) FAST NUCCS

UNIVERSITY

M.

Question #13(a) :-

Soln.

(i)

$$C \leq S$$

$$E \leq U$$

$$B \leq R$$

$$B \leq R$$

$$O \leq E$$

$$X \leq N$$

$$N \leq D$$

$$O \leq E$$

$$B \leq R$$

$$X \leq N$$

$$Y \leq O$$

$$G \leq W$$

(ii) $L \leq B$

$$O \leq E$$

$$W \leq M$$

$$I \leq Y$$

$$P \leq F$$

$$B \leq R$$

$$S \leq I$$

$$O \leq E$$

$$X \leq N$$

$$N \leq D$$

$\Rightarrow$ BE MY FRIEND

Ans

$\Rightarrow$ SURRENDER NOW

Ans

Question #14(a) :-

$$h(K) = K \bmod 97$$

(i) $034567981 \bmod 97$

Soln.

$$\Rightarrow 97 \times 356371 = 34567987$$

$$\text{difference} = 34567982 - 34567987 = -6$$

$$34567982 \div 97$$

$$97 \times 356371 = 34567987$$

$$97 \times 356000 = 34532000 \quad \text{remainder} = 35982$$

$$97 \times 371 = 35987 \quad \text{remainder} = -6$$

$$-6 \bmod 97 = 91, \text{ so } h = 91$$

Ans

imes Year
roll-no

Ans

Name: Yash - Raj

Roll-no: 24K-0737.

Question #14 (a) :-

(ii) $183211232 \bmod 97$

Soln.

$97 \times 1888775 = 18321175$

remainder 57. so $h=57$ Ans.

(iii) $220195744 \bmod 97$

Soln.

$97 \times 2270059 = 220195723$

remainder 21

 $h=21$ Ans.

(iv) $987255335 \bmod 97$

Soln.

$97 \times 10177890 = 987255330$

remainder 5

 $h=5$ Ans.Question #14 (b) :-

(i) $104578690 \bmod 101$

Soln.

$101 \times 1035383 = 104573683$, remainder 5007.

$5007 \bmod 101 \Rightarrow 101 \times 4949$, remainder 58

 $h=58$.

(ii) $43222187 \bmod 101$

Soln-

$$101 \times 4279427 = 43222187$$

remainder is 52

$h = 52$ Ans.

(iii) $372201919 \bmod 101$

Soln

$$101 \times 3685167 = 372201919$$

remainder is 52

$h = 52$ Ans.

(iv) $501338753 \bmod 101$

Soln

$$101 \times 4963750 = 501338750$$

remainder is 3

$h = 3$ Ans.

Question #14(c):

(i) $x_{n+1} = (4x_n + 1) \bmod 7, x_0 = 3$

Soln-

$$x_1 = (4(3) + 1) \bmod 7 = 6$$

$$x_2 = (4(6) + 1) \bmod 7 = 4$$

$$x_3 = (4(4) + 1) \bmod 7 = 6$$

$$x_4 = (4(3) + 1) \bmod 7 = 6$$

$$x_5 = (4(6) + 1) \bmod 7 = 4$$

Name: Yash.

Roll: 24K-

BCS-3H

Question

(ii)

Name: Yash-Ray

Roll: 24K-0737.

BCS-3H

Question #14(c):

$$(ii) \quad x_{n+1} = (7x_n + 2) \bmod 9, \quad x_0 = 5$$

$$\text{Soln.} \quad x_1 = (7(5) + 2) \bmod 9 = 1$$

$$x_2 = (7(1) + 2) \bmod 9 = 0$$

$$x_3 = (7(0) + 2) \bmod 9 = 2$$

$$x_4 = (7(2) + 2) \bmod 9 = 7$$

$$x_5 = (7(7) + 2) \bmod 9 = 6$$

Am

Question #15:

$$(a) (i) \quad d_{12} = [10 - (x(d_1 + d_3 + d_5 + d_{11}) + (d_2 + d_4 + \dots + d_{10})) \bmod 10] \bmod 10$$

Soln.

$$\text{Odd sum} = 81$$

$$\text{Even sum} = 14$$

$$\text{Total} = 81 + 14 = 95$$

$$95 \bmod 10 = 5$$

$$\text{check digit} = (10 - 5) \bmod 10 = 5$$

$$(ii) \quad \text{Odd sum} = 99$$

$$\text{Even sum} = 19$$

$$\text{Total} = 99 + 19 = 118$$

$$118 \bmod 10 = 8$$

$$\text{check digit} = (10 - 8) \bmod 10 = 2$$

Am

Question #15 (b):

(i) Odd sum = 42

Even sum = 18

$$42 + 18 = 60 \Rightarrow 60 \bmod 10 = 0 \text{ (valid)}$$

(ii)

odd sum = 60

Even sum = 28

$$\text{Total} = 60 + 28 = 88 \rightarrow 88 \bmod 10 = 8 \text{ (Not-valid)}$$

Question #15 (c):

$(2d_1 + 2d_2 + \dots + ad_9) \bmod 11$, if 10 then n:-

Sum

Digits: 0, 0, 7, 1, 1, 8, 8, 8, 7

$$\text{Sum} = 0 + 0 + 21 + 4 + 5 + 54 + 56 + 64 + 19$$

$$= 213 \bmod 11 = 4$$

ISBN = 0-321-50002-8

Digits: 0, 3, 2, 1, 5, 0, 0, 0, 1 Check-digit = 8

$$\text{Sum} = 0 + 6 + 6 + 4 + 25 + 0 + 0 + 89 + 9$$

$$(80 + 80) \bmod 11 = 8$$

$$50 \bmod 11 = 6 \quad 50, (6 + 80) \bmod 11 = 8$$

$$80 \bmod 11 = 2$$

2) Inverse of 8 mod 11 is 7 ($8 \times 7 = 56 \equiv 1$) so $Q = 7 \times 2 = 14 \equiv 3 \pmod{11}$

so Q is digit 0-9, $Q = 3$

James Yash
Roll-no: 1
Question
(a)

Name: Yash-Raj

Roll-no: 24K-0737.

Question #16 (a):

(a) Soln

$$p = 43, q = 59$$

$$n = 43 \times 59 = 2537$$

$$k = (43-1) \times (59-1) = 2494$$

$$c = 13$$

$$C = M^c \bmod n$$

$$P(AT) = 19^{13} \bmod 2537 \Rightarrow$$

$$AT = M = 0 \times 100 + 19 = 19$$

$$C(TA) = 1900^{13} \bmod 2537 \Rightarrow$$

$$TA = M = 19 \times 100 + 66 = 1966$$

$$C(CK) = 210^{13} \bmod 2537 \Rightarrow$$

$$CK = M = 2 \times 100 + 10 = 210$$

(b) Soln

$$n = 13 \times 19 = 247, c = 47$$

$$C = M^c \bmod n$$

Assignment

$$C(A) = 0^{47} \bmod 247$$

$$C(S) = 18^{47} \bmod 247$$

$$C(S) = 18^{47} \bmod 247$$

$$C(I) = 8^{47} \bmod 247$$

$$C(G) = 6^{47} \bmod 247$$

$$C(N) = 13^{47} \bmod 247$$

$$C(M) = 12^{47} \bmod 247$$

$$C(E) = 4^{47} \bmod 247$$

$$C(N) = 13^{47} \bmod 247$$

$$C(T) = 19^{47} \bmod 247$$

Semester

$$C(S) = 18^{47} \bmod 247$$

$$C(E) = 4^{47} \bmod 247$$

$$C(M) = 12^{47} \bmod 247$$

$$C(E) = 4^{47} \bmod 247$$

$$C(S) = 18^{47} \bmod 247$$

$$C(T) = 19^{47} \bmod 247$$

$$C(E) = 4^{47} \bmod 247$$

$$C(R) = 17^{47} \bmod 247$$

Question #16(c):

(i) $5^{2003} \pmod{7}$

Soln ~

$$5^{79} \equiv 1 \pmod{7}$$

$$5^6 \equiv 1 \pmod{7}$$

$$a = dq + r$$

$$2003 = 333(6) + 5$$

$$5^{333 \times 6 + 5} \pmod{7}$$

$$5^5 \pmod{7} = 3 \text{ Ans.}$$

(iii) $5 \pmod{13}$

Soln $5^{12} \equiv 1 \pmod{13}$

$$5^{154 \times 12 + 1} \pmod{13}$$

$$(5^{12})^{2003} 5 \pmod{13}$$

$$5 \pmod{13} = 5 \text{ Ans.}$$

(ii) $5^{2003} \pmod{11}$

Soln

$$5^{10} \equiv 1 \pmod{11}$$

$$2003 = (200)10 + 3$$

$$5^{200 \times 10 + 3} \pmod{11}$$

$$5^3 \pmod{11} = 4 \text{ Ans.}$$

(iv) $5^{2003} \pmod{17}$

Soln ~

$$5^{16} \equiv 1 \pmod{17}$$

$$2003 = (125)16 + 3$$

$$(5^{16})^{125} 5^3 \pmod{17}$$

$$5^3 \pmod{17} = 6 \text{ Ans.}$$

Discrete Assignment 03

14

ames Yash-Raj

Roll no = 24K-0737.

Question #18(a) ::

$$x = 2 \pmod{7}$$

$$x = 3 \pmod{17}$$

$$x = 5 \pmod{19}$$

Soln

Use CRT:

$$m = 7 \times 17 \times 19 = 2261$$

$$m_1 = 2261/7 = 323$$

$$m_2 = 133, m_3 = 119$$

$$y_1 = 323^{-1} \pmod{7}$$

$$1 = 1(323) + (-46)7$$

$$\boxed{y_1 = 2}$$

$$y_2^{-1} = 133 \pmod{17}$$

$$133 = 7(17) + 14$$

$$17 = 1(14) + 3$$

$$14 = 4(3) + 2$$

$$3 = 1(2) + 1$$

$$\boxed{y_2^{-1} = -6 + 17 = 11}$$

$$y_3^{-1} = 119 \pmod{19}$$

$$119 = 6(19) + 5$$

$$19 = 3(5) + 4$$

$$5 = 1(4) + 1$$

$$\boxed{y_3^{-1} = 4}$$

$$(2 \times 323 \times 1) + (3 \times 133 \times 11) + (5 \times 119 \times 4) \pmod{2261}$$

$$= 7415 \pmod{2261} = 632$$

$$(b) f(p) = (pth) \pmod{26}$$

Soln

$$f(s) = (18 + 3) \pmod{26} = 21 = v$$

$$f(i) = n$$

$$f(x) = a$$

$$f(h) = k$$

$$f(u) = x$$

$$f(n) = Q$$

$$f(d) = g$$

$$f(r) = h$$

$$f(e) = h$$

$$f(d) = g$$

$$f(a) = d$$

$$f(n) = Q$$

$$f(d) = g$$

$$f(T) = L$$

$$f(r) = h$$

$$f(t) = w$$

$$f(y) = b$$

$$f(f) = w$$

$$f(w) = z$$

$$f(o) = v$$

$$\Rightarrow \text{VNA KXQG-UHG DQG LUWRWZR}$$

Ans

$$(c) x^{302} \pmod{7}$$

Soln

$$x^6 \equiv 1 \pmod{7}$$

$$302 = 50(6) + 2$$

$$(11^6)^{50} \times 11^2 \pmod{7} = 2$$

Ans

Question #18 :-

(a) soln

No: of floor = 27

offices per floor = 37

Using product rule :-

Total offices = no: of floor $\times$ offices per floor = 999 Ans.

(b) soln

Colors = 12

Sizes = 3

various = 2.

Combining sum & product.

Total types = $(12 \times 3) + (12 \times 3) = 72$ Ans

(c) different three letter initials.

$26 \times 25 \times 24 = 15600$ Ans.

Question #19 :-

(a) 5 players to select from 13 girls (each girl can play each position)
soln doesn't matter so,

$${}^nC_5 = {}^{13}C_5 = \frac{13!}{(13-5)!5!} = 1287 \text{ Ans.}$$

(b) soln ${}^nP_5 = {}^{75}P_5 \Rightarrow \frac{75!}{(75-5)!} \Rightarrow \frac{75 \times 74 \times 73 \times 72 \times 71 \times 70!}{70!} = 2072126800$ Ans

(c) soln ${}^nC_4 = {}^{27}C_4 = \frac{27!}{23! \cdot 4!} = 17550$ Ans

no = Yash-R
no = d

Justic

Yash-Raj

no: d4k-0737.

Question #20 (a):

Soln

Hexa-decimal-digits = 16.

lengths 10, 26 or 58.

all WEP Keys = $16^{10} + 16^{26} + 16^{58}$ Ans

(b) Soln-

String (length 4) with letter n:-

Total 4 length string = 26^4 .4 length strings without letter 'n' = 25^4

Strings with letter n:-

 $26^4 - 25^4 = 66,351$ Ans

(c) Soln-

Initials with no repetition:-

 $26 \times 25 \times 24 \Rightarrow 15600$ AnsQuestion #21:(a) Soln: Total functions = $(2 \times 2 \times 2 \dots \times 2)$ (m times).T.F = 2^m Ans.(b) Soln: $n = m = 5$ So, T.F = $(5)(5-1)(5-2)(5-3)(5-4)$.T.F = $5 \times 4 \times 3 \times 2 \times 1 = (120)$ Ans

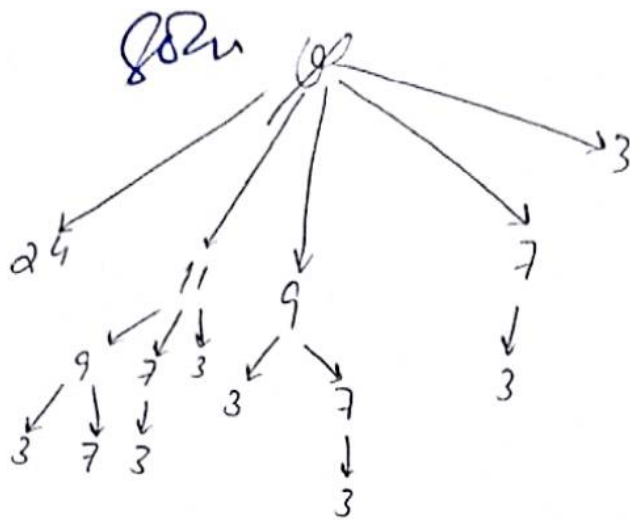
(c) Soln-

 $h = 15, e = 48, n = 24, m = 34, Cn = 28, ck = 28$. $15 \times 48 \times 24 \times 34 \times 28 \times 28$

= 1460615680 Ans

Question #22:

(a) $\{3, 7, 9, 11, 24\}$ sum 228.



∴ Total subsets: $\{3, 24\}, \{24\}, \{11\}, \{9\}, \{7\}, \{3\}, \{11, 9, 3\}, \{11, 7, 3\}, \{11, 3\}, \{9, 3\}, \{9, 7, 3\}, \{7, 3\}, \{7, 9\}, \{7, 11\}, \{9, 11\}$

(b) 802

0000, 0001, 0010, 0100, 1000, 1001, 1001, 0101

802 8 strings Ans.

(c) $\{A, B, C, D\}$ P, T, S.

802.

- A can't be president, C or D must be secretary.
- Can I = (secretary of C).
 - 2 choices president (B or D).
 - $2 \times 2 = 4$ Ans.
- Can 2 = (secretary of D).
 - 2 choices of president (B or C).
 - $2 \times 2 = 4$

Total: $4 + 4 = 8$ Ans

Yash Raj

no: 24K-0737.

Question #23 :-

(a) Soln - Total numbers = 8
partitions = 2

$$\therefore nC_3 = \frac{8!}{3!5!} \Rightarrow 56 \text{ ways Ans.}$$

(b) Soln

$$nC_4 = \frac{12!}{4!8!} \Rightarrow 35 \text{ ways Ans.}$$

(c) Soln

$$nC_5 = \frac{9!}{5!4!} \Rightarrow 126 \text{ ways Ans.}$$

Question #24 :-

(a) Soln 20 people
5 different partitions.

$${}^{20}P_5 = \frac{20!}{15!} \Rightarrow 1860480 \text{ ways Ans.}$$

(b) Soln

$${}^{16}P_4 = \frac{16!}{12!} \Rightarrow 43680 \text{ ways Ans.}$$

(c) Soln

$${}^{15}P_2 = \frac{15!}{13!} \Rightarrow 210 \text{ ways Ans.}$$

Question #25

(a)

$$\text{Soln } {}^5C_1 \times {}^3C_2 \times {}^4C_1 \times {}^6C_3$$

$$\Rightarrow 5 \times 3 \times 4 \times 20 = 1200 \text{ Ans.}$$

(b)

$$\text{Soln } {}^nC_1 \Rightarrow {}^{12}C_5 \Rightarrow 792 \text{ Ans.}$$

(c)(i) Begin with 4:

$$(i) \text{ With repetition: } 1 \times 10 \times 10 \times 10 \times 10 \Rightarrow 10000$$

$$(ii) \text{ Without: } 1 \times 9 \times 8 \times 7 \times 6 \Rightarrow 3024 \text{ Ans.}$$

(ii) Begin with 3 or 2

$$(i) \text{ With repetition: } 2 \times 10 \times 10 \times 10 \times 10 \Rightarrow 20000$$

$$(ii) \text{ Without: } 2 \times 9 \times 8 \times 7 \times 6 \Rightarrow 6048 \text{ Ans.}$$

Question #26 :-

(a) length = 10

$$\text{Soln } |A| = 2^7, |B| = 2^8$$

By inclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cap B| = 2^5$$

$$(A \cup B) = 2^7 + 2^8 - 2^5 \Rightarrow 352$$

Ans

(b) length = 5

Soln

$$|A| = 2^4, |B| = 2^3$$

$$|A \cap B| = 2^2$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 20$$

Ans

me: Yash - Raj

no: 24K-0737.

CS-3H.

Question #26(c) :-

(i) 2 letters followed by 4 digits:-

soln
with repetition = $26 \times 26 \times 10 \times 10 \times 10 \times 10 \Rightarrow 6760000$ Ans

without repetitions $26 \times 25 \times 10 \times 9 \times 8 \times 7 \Rightarrow 327600$ Ans

(ii) 3 odd digits followed by 2 vowels:-

soln
with repetition = $5 \times 5 \times 5 \times 5 \times 5 \Rightarrow 3125$

without repetitions $5 \times 4 \times 3 \times 5 \times 4 \Rightarrow 1200$ Ans

Question #27 :-

(a) soln. 26 possible letters
with 30 students

$$\left\lceil \frac{30}{26} \right\rceil = 2 \text{ student, } \text{Ans}$$

(b) soln ~~26 possible letters with 261~~ To prove at least 9
people have 1000000 hairs.
Total population = 8008278

$$\left\lceil \frac{8008278}{1000000} \right\rceil = 9 \text{ Ans}$$

(c) soln
time-periods = 38
total-classes = 677

$$\left\lceil \frac{677}{38} \right\rceil = 18 \text{ Ans}$$

Question #18(a):

(i) x^5 in $(1+x)^n$

Soln
 $(x+y)^n = \sum_{j=0}^n \binom{n}{j} x^j y^{n-j}$
 here $x=1, y=x$

$\binom{n}{5} = \frac{n!}{5!(n-5)!}$
Ans

(ii) $a^7 b^{17}$ in $(a-b)^{24}$

Soln
 $n=24, n-j=7 \Rightarrow j=17$
 $n-j=7$
 $j=24-7=17$

$\binom{24}{17} (a)^7 (-1)^{17} = (346104)(-1) = \boxed{-346104}$
Ans

part(b):

(i) Soln

			1		
		1		1	
	1	2	1		
1	3	3	1		
1	4	6	4	1	
1	5	10	10	5	1

Ans

(ii) Expand $(3x-2)^4$ using Pascal's.

Soln
 $n=4$

1 4 6 4 2

$1(a)^4(b^0) = 81x^4$

$4(a^3)(b^1) = -216x^3$

$6(a^2)(b^2) = 216x^2$

$4(a^1)(b^3) = -96x$

$1(a^0)(b^4) = 16$

c) $81x^4 - 216x^3 + 216x^2 - 96x + 16$

Ans

Name: Yash Raj

Roll-no: 2416-0737.

BLS-34.

Question # 29

(a) Soln

20 women

16 men

total = 36

(i) $36!$ Ans(ii) ${}^{36}P_7 = 42672307200$ Ans

(iii) all women on left & all men on right.

 $[20 \text{ women}] [16 \text{ men}]$ $\downarrow$
 $20!$ $\downarrow$
 $16!$ Ans(b) $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$ Ans(c) ${}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3$
 $= 299$ Ans(c) (i) $(2x+y)^6$

$$(a+b)^n = \sum_{j=0}^n \binom{n}{j} a^{n-j} b^j$$

$$= \binom{6}{0} (2x)^6 + \binom{6}{1} (2x)^5 (y) + \binom{6}{2} (2x)^4 (y^2) + \binom{6}{3} (2x)^3 (y^3) + \binom{6}{4} (2x)^2 (y^4) +$$

$$\binom{6}{5} (2x) (y^5) + \binom{6}{6} (y^6)$$

$$= 64x^6 + 192x^5y + 240x^4y^2 + 160x^3y^3 + 60x^2y^4 + 12xy^5 + y^6$$

$$(ii) (3a - 2b)^5$$

Soln

$$\binom{5}{0}(3a)^5 + \binom{5}{1}(3a)^4(-2b) + \binom{5}{2}(3a)^3(-2b)^2 + \binom{5}{3}(3a)^2(-2b)^3 + \binom{5}{4}(3a)(-2b)^4 + \binom{5}{5}(-2b)^5$$

$$\binom{5}{5}(-2b)^5$$

$$= 243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$$

Question #30 :-

$$(a) \text{ Soln } {}^{15}C_3 \times {}^{12}C_3 \times {}^7C_3 \times {}^5C_3 \times {}^9C_3 \times {}^{10}C_3$$

$$= 455 \times 220 \times 35 \times 10 \times 84 \times 120$$

$$= 353152800000$$

(b) Soln.

Students = 97.

positions - grades = 10.

$$\lceil 97/10 \rceil = 10$$

(c) Soln

Squad = 15 players.

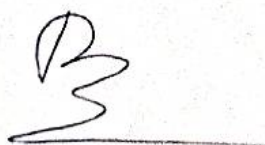
w.k.s = 2, bowlers = 5, all-rounders = 3, batsmen = 5

Captain choice = 15

$$v-c \Rightarrow 15 - 1 = 14.$$

w.k.s = 2

$$15 \times 14 \times 2 = 420$$



Name:- Yash-Raj
Roll-no:- 24k-0737.
BLS-3H.

Question #31:-

(a) Combinations:-

- (i) lottery ticket selection
- (ii) teams from group.

(b) permutations:-

- (i) passwords
- (ii) order of cars.

(c) Binomial theorem:-

- (i) binomial expansion
- (ii) Expanding expressions

(d) proof-method:-

- (i) verifying correctness of algo.
- (ii) ensuring consistency in software.

(e) Mathematical inductions:-

- (i) proving formulae.
- (ii) Validating correctness of recursive algo (steps).

Ans.